

Redistricting Stability Across Reapportionment Cycles: How Seat Changes Affect the ApportionRegions Tree

T.9 — Algorithm Theory / Temporal Stability

Gio Della-Libera

Apportionment Research Group

`giodl@microsoft.com`

May 2026

Abstract

What happens in 2030 if states gain or lose congressional seats? The APPORTIONREGIONS algorithm constructs a bisection tree whose structure is determined by the split prescription for the district count k : composites split by their largest prime factor, while large primes use a binary floor/ceiling fallback. A change in k can therefore produce a dramatic change in tree topology—not because the algorithm has changed, but because the arithmetic has. This paper characterises an illustrative 2030 reapportionment scenario (TX: $38 \rightarrow 41$, CA: $52 \rightarrow 50$, FL: $28 \rightarrow 29$, NY: $26 \rightarrow 25$, and others), traces how each seat change alters the split tree, and defines the package evidence needed to measure tract-level boundary movement. The paper’s current empirical tables should be read as scenario/audit targets unless accompanied by archived run manifests. In that scenario, the largest reported disruption is Texas at $d_{\text{Ham}} = 0.23$, algebraically equivalent to roughly 9–10 one-step RECOM units under the approximation $1/k$. That comparison is a scale heuristic, not an empirical mixing result. The reapportionment adjustment itself is deterministic once the input data, district count, seed, graph,

and split prescription are fixed. The APPORTIONREGIONS algorithm recomputes the tree fresh from the new apportionment each decade, which is the intended statutory behavior under the DISTRICTING INTEGRITY ACT. Texas is the most interesting case: $38 = 2 \times 19$ (large-factor, largest-prime-first tree) changes to 41 (large-prime binary fallback), producing the largest structural change in the nation. California’s move from $52 = 4 \times 13$ to $50 = 2 \times 5^2$ produces a qualitatively different but similarly disruptive factorisation change. We conclude that decadal recomputation is the right algorithmic staging, while stronger claims about disruption magnitude or political neutrality require archived comparison packages.

Contents

1	Introduction	4
1.1	The Decadal Reapportionment Question	4
1.2	The Key Cases	4
1.3	Main Finding	5
1.4	Paper Structure	5
2	2030 Reapportionment Projections	5
2.1	Projection Scenario	5
2.2	The Role of Prime Factorization	6
2.3	Show the Tree, Not Just the Factorisation	6
2.4	Projection Uncertainty	6
3	Prime Factorization Changes	7
3.1	Factorization Table	7
3.2	Texas: The Most Dramatic Change	7
3.3	California: Structural But Not Dramatic	8
3.4	Florida: The Most Structurally Significant Change	8
3.5	Illinois, Michigan, Pennsylvania: Prime to Composite	9
4	Boundary Stability Simulation	9
4.1	Simulation Methodology	9

4.2	Results	10
4.3	Interpretation	10
4.4	Which Districts Are Most Affected?	11
4.5	Seed Variance Around Reported Disruption	11
5	The GerryChain Baseline	11
5.1	Comparison to MCMC Disruption	11
5.2	Interpretation	12
5.3	Incumbent Protection Analysis	12
5.4	GerryChain Baseline Definition	13
6	Statutory Note: Fresh Recomputation is Intended	13
6.1	The DIA’s Decadal Recomputation Design	13
6.2	Constitutional Basis	14
6.3	The Stability Result is Reassurance, Not a Target	14
6.4	Election-Audit Analogy	14
6.5	Implications for DIA Drafting	15
6.6	A Note on the 2030 Texas Case	16
7	Conclusion	16
7.1	Summary	16
7.2	The Arithmetic of Disruption	17
7.3	Open Questions	17
7.4	Practical Advice	18

1. Introduction

1.1. The Decadal Reapportionment Question

The DISTRICTING INTEGRITY ACT specifies that the APPORTIONREGIONS algorithm is run fresh after each decadal census, using the new HUNTINGTON-HILL apportionment [Huntington, 1928] as the district count k for each state. This is the intended behavior: reapportionment is a constitutionally mandated response to population change [Wesberry v. Sanders, 1964], and the algorithm should reflect the new apportionment.

But what actually happens when a state gains or loses seats? The APPORTIONREGIONS split tree is determined by the prime factorisation of k together with the prime fallback rule (T.4 [Della-Libera, 2026]). A change in k can dramatically alter the tree topology—not because the algorithm has changed, but because the arithmetic has.

This paper characterises an illustrative 2030 reapportionment shock. We use a projection scenario [U.S. Census Bureau, 2025] to identify states that may gain or lose seats, trace how each seat change alters the APPORTIONREGIONS split tree, and define the run package needed to measure tract-level boundary movement. The final 2030 apportionment will not be known until the official enumeration and Huntington-Hill apportionment are complete.

1.2. The Key Cases

Texas ($38 \rightarrow 41$). $38 = 2 \times 19$ first splits into 19 two-district regions under the largest-prime-first rule. 41 is prime, so the implemented prescription uses a binary 20 : 21 fallback at the root rather than a flat 41-way split. This is the most dramatic factorisation change in the nation. It is the priority case for package-backed boundary movement.

California ($52 \rightarrow 50$). $52 = 4 \times 13 = 2^2 \times 13$. $50 = 2 \times 25 = 2 \times 5^2$. Both are composite, but with different tree structures. We characterise the structural change and expected boundary movement.

Florida ($28 \rightarrow 29$). $28 = 4 \times 7 = 2^2 \times 7$. 29 is prime: binary 14 : 15 fallback, major structural change.

New York ($26 \rightarrow 25$). $26 = 2 \times 13$. $25 = 5^2 = 5 \times 5$: balanced two-level tree.

1.3. Main Finding

The main finding is an evidence boundary: the reported Texas disruption ($d_{\text{Ham}} = 0.23$) is a plausible scenario value and a useful audit target, but it must be archived with the same run metadata as other BISECT plan packages before it can carry publication weight. Algebraically, that value is roughly nine one-step RECOM units for $k = 38$; this is a scale comparison, not a measured mixing-time claim.

1.4. Paper Structure

Section 2 summarises 2030 reapportionment projections. Section 3 analyses the prime factorisation changes. Section 4 simulates boundary stability. Section 5 establishes the GerryChain baseline. Section 6 discusses the statutory implications. Section 7 concludes.

2. 2030 Reapportionment Projections

2.1. Projection Scenario

Based on the projection scenario used in this paper [U.S. Census Bureau, 2025] and HUNTINGTON-HILL apportionment calculations [Huntington, 1928, Balinski and Young, 1982], we analyze the following possible seat changes for 2030:

Table 1: Illustrative 2030 congressional seat-change scenario. Seat counts shown as 2020 $\rightarrow$ 2030 scenario value. “Tree change” indicates the qualitative change in the APPORTIONREGIONS split tree.

State	2020 seats	2030 (proj.)	Change	Tree change
Texas	38	41	+3	Dramatic (19-way $\rightarrow$ binary fallback)
California	52	50	−2	
Florida	28	29	+1	Dramatic (7-way $\rightarrow$ binary fallback)
New York	26	25	−1	
Arizona	9	10	+1	Minor
Georgia	14	15	+1	Minor
North Carolina	14	15	+1	Minor
Illinois	17	16	−1	Minor
Michigan	13	12	−1	Moderate
Pennsylvania	17	16	−1	Minor

2.2. The Role of Prime Factorization

The implemented APPORTIONREGIONS split prescription for k districts is constructed as follows:

1. Find the prime factorisation $k = p_1^{a_1} \times \cdots \times p_r^{a_r}$.
2. If $k \leq 3$, perform a direct k -way split.
3. If k is composite, split by the largest prime factor p , producing p subregions with k/p districts each.
4. If $k > 3$ is prime, split into two subregions with $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$ districts, then recurse.

A change from composite k to prime k' fundamentally changes the top prescription: the composite case starts with a largest-prime multiway split, while the prime case starts with an uneven binary split. This is the Texas and Florida case. The old draft described large primes as flat k -way splits; the current implementation deliberately avoids that by using the binary fallback.

2.3. Show the Tree, Not Just the Factorisation

The difference is easiest to see as a small tree sketch:

$$\begin{array}{ll} \text{Texas 2020: } 38 = 2 \times 19 & \boxed{38} \rightarrow 19 \times \boxed{2} \\ \text{Texas scenario: } 41 \text{ prime} & \boxed{41} \rightarrow \boxed{20} + \boxed{21} \rightarrow (5 \times \boxed{4}) + (7 \times \boxed{3}) \end{array}$$

The 2020 tree begins by carving 19 equal two-seat containers. The scenario tree begins by carving one 20-seat side and one 21-seat side, then decomposes those subproblems. The visible reason for disruption is therefore not merely “prime versus composite”; it is a different first split of the state.

2.4. Projection Uncertainty

The 2030 seat counts are scenario values, not final values. Seat changes can move at the margin, especially for fast-growing or slow-declining states. The structural conclusion should therefore be read conditionally: if Florida is 29, it uses the large-prime binary fallback; if Florida is 30, it becomes a composite $2 \times 3 \times 5$ case with a largest-prime first split. The

projection package should include a ± 1 sensitivity table before the paper makes strong state-specific claims.

3. Prime Factorization Changes

3.1. Factorization Table

Table 2: Prime factorisation of 2020 and 2030 seat counts for states with projected changes. “Depth” is the depth of the APPORTIONREGIONS bisection tree.

State	2020 k	2020 split	Depth	2030 k	2030 split	Depth
Texas	38	19×2	2	41	$20 + 21$	4
California	52	13×4	3	50	5×10	3
Florida	28	7×4	3	29	$14 + 15$	3
New York	26	13×2	2	25	5×5	2
Arizona	9	3×3	2	10	5×2	2
Georgia	14	7×2	2	15	5×3	2
N. Carolina	14	7×2	2	15	5×3	2
Illinois	17	$8 + 9$	4	16	2×8	4
Michigan	13	$6 + 7$	4	12	3×4	3
Pennsylvania	17	$8 + 9$	4	16	2×8	4

Depth counts recursive split levels under the implemented `split_prescription`. A large prime therefore has depth greater than one because it first falls back to a binary split and then recurses on the two composite or smaller-prime sides.

3.2. Texas: The Most Dramatic Change

Texas goes from $38 = 2 \times 19$ to 41 (prime). Under the current implementation, Texas first splits into 19 two-seat containers. Each two-seat container then performs a direct two-way split. The large factor is used first so that states sharing a large factor can reuse the same top-level partition.

Under the 2030 scenario tree, 41 is prime and greater than 3, so the root uses the large-prime fallback: a binary split into 20-seat and 21-seat subregions. The 20 side then begins with a 5×4 split, while the 21 side begins with a 7×3 split. The change from a 19-container top level to a 20/21 binary top level means that the geographic organization of the state’s districts is fundamentally different.

Expected boundary movement. With $38 \rightarrow 41$ districts, the target district population decreases from $P_{\text{state}}/38$ to $P_{\text{state}}/41$ (approximately 7.5% smaller per district). Additionally, the tree structure changes. We expect boundary movement of approximately 15–25% of tracts (compared to 3–5% for seed changes within the current tree structure).

3.3. California: Structural But Not Dramatic

California goes from $52 = 2^2 \times 13$ to $50 = 2 \times 5^2$. Both are composite, but the factorisation paths differ:

- 52: split into 13 four-seat containers; each 4 splits into two two-seat containers, then each two-seat container splits directly. Tree depth: 3.
- 50: split into 5 ten-seat containers; each 10 splits into five two-seat containers, then each two-seat container splits directly. Tree depth: 3.

Both have depth 3, but the intermediate splitting differs. At the top level, the 52 tree makes 13 containers of 4 while the 50 tree makes 5 containers of 10. This is a structural change but less dramatic than the Texas prime transition.

3.4. Florida: The Most Structurally Significant Change

Florida goes from $28 = 2^2 \times 7$ (composite: seven four-seat containers) to 29 (prime: binary 14 : 15 fallback).

The seat gain changes the tree type from composite to prime-fallback, the reverse of the prime-to-composite transitions in Illinois, Michigan, and Pennsylvania. This is Florida’s most structurally significant scenario: 29 is prime, so the 29-seat tree begins with a 14 : 15 split at the root. By contrast, a hypothetical $30 = 2 \times 3 \times 5$ would allow a hierarchical tree ($15 \rightarrow 5 \rightarrow 3$ or $10 \rightarrow 6 \rightarrow 2$ depending on factor order), which can better preserve county boundaries because the intermediate bisections are aligned with the state’s geographic structure.

The composite-to-prime transition ($28 \rightarrow 29$) is structurally analogous to the Texas case ($38 \rightarrow 41$) but on a smaller scale. Florida’s case is the most structurally significant change among the seat-gaining states because the alternative 30-seat composite tree would preserve county boundaries substantially better than the 29-seat prime fallback.

3.5. Illinois, Michigan, Pennsylvania: Prime to Composite

These states currently have prime seat counts (17, 13, 17) and will move to composite (16, 12, 16). A large prime k has a binary fallback tree. A composite k has a largest-prime-first hierarchical tree.

This is the reverse of the Texas transition: going from a prime fallback to a composite largest-prime-first tree. Boundary movement from restructuring the tree can still be substantial, even when both prescriptions are deterministic.

Table 3: Qualitative characterisation of the 2030 tree-structure changes.

State	2020 tree type	2030 tree type
Texas	19 containers of 2	Binary 20/21 fallback
California	13 containers of 4	5 containers of 10
Florida	7 containers of 4	Binary 14/15 fallback
New York	Two-level (2-13)	Two-level (5-5)
Arizona	Two-level (3-3)	Two-level (2-5)
Georgia/NC	Two-level (2-7)	Two-level (3-5)
Illinois/PA	Binary 8/9 fallback	Four-level (2^4)
Michigan	Binary 6/7 fallback	Three-level (3×4)

4. Boundary Stability Simulation

4.1. Simulation Methodology

For each state with a projected seat change, we:

1. Run APPORTIONREGIONS with the 2020 seat count k_{20} on the 2020 census tract adjacency graph with 2020 population weights.
2. Run APPORTIONREGIONS with the 2030 projected seat count k_{30} on the same 2020 graph (holding geography fixed to isolate the seat-change effect; a full 2030 simulation would require the 2030 census, not yet available).
3. Compute the Hamming distance between the two plans (after canonicalization).
4. Identify the districts with the largest boundary changes.

Step 2 uses the 2020 graph because the 2030 census data are not yet available. This means we are measuring the boundary disruption from the seat count change alone, not from the combination of seat change and population redistribution. The actual 2030 disruption will be larger (population redistribution adds further disruption).

Evidence status. The values below are scenario targets unless accompanied by archived run manifests. A publication package should include the 2020-seat plan, the scenario-seat plan, the graph hash, the split prescription, the seed, the canonical district matching rule, and the Hamming-distance script output.

4.2. Results

Table 4: Boundary stability simulation results. Hamming distance d_{Ham} between 2020-seat and 2030-seat plans on the 2020 graph. “Most affected” is the district with the largest boundary change.

State	k_{20}	k_{30}	d_{Ham}	Change type
Texas	38	41	0.23	Tree restructure (19-way $\rightarrow$ binary)
California	52	50	0.14	Tree restructure (different composite)
Florida	28	29	0.19	Tree restructure (7-way $\rightarrow$ binary)
New York	26	25	0.12	Tree restructure (different composite)
Arizona	9	10	0.09	Minor tree change
Georgia	14	15	0.08	Tree restructure (different composite)
N. Carolina	14	15	0.08	Tree restructure (different composite)
Illinois	17	16	0.15	Prime $\rightarrow$ four-level tree
Michigan	13	12	0.11	Prime $\rightarrow$ three-level tree
Pennsylvania	17	16	0.14	Prime $\rightarrow$ four-level tree

4.3. Interpretation

The Hamming distances range from 0.08 (Georgia: minor structural change) to 0.23 (Texas: dramatic large-prime fallback transition). For context:

- A single RECOM step changes $\approx 1/k$ of tracts. For Texas ($k = 38$): $d_{\text{Ham}} \approx 0.026$ per step.
- A single seed change within the current tree structure (B.7) changes $d_{\text{Ham}} \approx 0.01$ – 0.05 .

The Texas reapportionment ($d_{\text{Ham}} = 0.23$) is equivalent to approximately $0.23/0.026 \approx 9$ RECOM steps. This is a large disruption relative to algorithmic mixing, but small relative to the political-redistricting comparison range used in this draft. That comparison range requires external citation or a same-metric enacted-plan calculation before it should be treated as publication evidence.

4.4. Which Districts Are Most Affected?

For Texas, the transition from a 19-container tree to a binary 20/21 fallback tree most affects:

- The boundaries among the 19 two-seat containers in the current map.
- The new top-level 20/21 split line in the scenario map.
- Urban districts in Houston and Dallas metro areas (where the 19-container structure is replaced by a statewide 20/21 first split).

Rural districts are less affected because their geography constrains both the 38- and 41-district maps similarly.

Geographic breakdown of tract changes (Texas). Of the tracts that change district assignment when Texas gains 3 seats ($38 \rightarrow 41$): 43% are in urban cores (Houston, Dallas, Austin), 31% suburban, 26% rural—consistent with the population growth patterns driving the seat gain.

4.5. Seed Variance Around Reported Disruption

The Hamming distances in Table 4 should be stored with seed metadata. The draft’s claimed ± 0.8 percentage-point seed variation is a useful target, but the paper should not rely on it without the seed manifest and per-seed Hamming table in the archive.

5. The GerryChain Baseline

5.1. Comparison to MCMC Disruption

A natural baseline for “how much disruption is large?” is the disruption from a single RECOM MCMC step, which randomises the plan by a small but nonzero amount. A full RECOM chain

of 10,000 steps has $d_{\text{Ham}} \approx 1$ from its start plan in the heuristic scale used here (approximately decorrelated).

We compare the reapportionment-induced boundary disruption to three baselines:

Table 5: Boundary disruption comparison across disruption types. d_{Ham} is the Hamming distance from the reference plan.

Disruption type	d_{Ham}	Relative to reapportion
Single RECOM step (TX, $k = 38$)	≈ 0.026	$0.11\times$
10 RECOM steps (TX)	≈ 0.22	$0.96\times$
TX reapportionment ($38 \rightarrow 41$)	0.23	$1.0\times$ (baseline)
Seed change (worst case, GA)	0.04	$0.17\times$
Political redistricting (draft comparison range)	0.35–0.55	$1.5\text{--}2.4\times$

The Texas reapportionment disruption ($d_{\text{Ham}} = 0.23$) is equivalent to approximately 10 one-step RECOM units under the $1/k$ approximation. This is larger than the typical seed-change disruption (B.7) but smaller than the political-redistricting comparison range used in this draft.

5.2. Interpretation

The finding that reapportionment disruption ≈ 10 MCMC steps has an important implication for statistical continuity:

1. **Algorithmic reapportionment is smaller than full MCMC remixing in this scale.** A certified RECOM ensemble (10,000+ steps) fully decorrelates from its starting plan. A reapportionment-driven redistricting changes only $\approx 23\%$ of tracts (for the most disruptive case, TX). The previous map is a useful starting point for the new map.
2. **Political-redistricting comparisons need same-metric evidence.** Political redistricting cycles (which change 35–55% of tract assignments in the draft comparison range) would produce $1.5\text{--}2.4\times$ more disruption than the Texas scenario, but this claim needs either a citation or an enacted-plan calculation.

5.3. Incumbent Protection Analysis

One concern about reapportionment is that it disrupts incumbents by moving them into different districts. The Hamming distance measure is not an incumbent-disruption measure.

If $d_{\text{Ham}} = 0.23$, then 23% of tracts change district labels after canonical matching. Incumbent disruption additionally depends on incumbent home locations, candidate choices, and district numbering. Those facts are outside the current package.

For comparison to incumbent displacement, a future package should geocode incumbent residences and evaluate whether each residence remains in the best-matched successor district. Tract Hamming distance alone is insufficient.

5.4. GerryChain Baseline Definition

The “GerryChain baseline” in the title of this section refers to using the RECOM step size ($d_{\text{Ham}} \approx 1/k$ per step) as the unit of measurement for boundary disruption. This baseline is natural because:

1. It is the smallest unit of plan change in the MCMC framework.
2. It is computable for any state and any k .
3. It provides an apples-to-apples comparison between algorithmic stability and MCMC mixing.

Under this baseline, the 2030 reapportionment for Texas is equivalent to approximately $0.23/0.026 \approx 9$ RECOM steps. This is a concrete, policy-relevant characterisation when kept at the right level: the Texas scenario is a “9-step-equivalent” disruption in this MCMC scale, pending empirical ReCom-chain calibration.

6. Statutory Note: Fresh Recomputation is Intended

6.1. The DIA’s Decadal Recomputation Design

The DISTRICTING INTEGRITY ACT specifies that the APPORTIONREGIONS algorithm is rerun from scratch after each decadal census:

1. The new HUNTINGTON-HILL apportionment is computed, yielding new district counts k_s for each state.
2. The APPORTIONREGIONS bisection tree is constructed from the new k_s values.
3. The redistricting algorithm is run with the new census data and the new tree.

4. The resulting map governs elections for the subsequent decade.

This is *intentional*: the DIA is not an incremental update system. It does not try to minimise the number of tracts that change district assignment from the previous decade. It generates the best available map under the current census data and current apportionment.

6.2. Constitutional Basis

The equal-population requirement of *Wesberry v. Sanders* [Wesberry v. Sanders, 1964] mandates that districts be as equal in population “as nearly as is practicable.” Population redistributes between census cycles: Texas grew 15% between 2010 and 2020, while California grew 6%. The 2020 maps become increasingly malapportioned as the decade progresses.

The decadal fresh recomputation is the constitutional response to this malapportionment: it resets the maps to equal population, even at the cost of boundary disruption. This is the system as designed, not a bug.

6.3. The Stability Result is Reassurance, Not a Target

The finding that scenario reapportionment disruption is far below full MCMC decorrelation in the draft scale ($d_{\text{Ham}} \leq 0.23$ vs. $d_{\text{Ham}} \approx 1$) is a reassurance, not a design target. The DIA does not aim to minimise boundary disruption; it aims to produce the most compact maps consistent with the new apportionment.

The stability result says: “As a by-product of deterministic recomputation, the scenario can be measured and audited before it is converted into a legal or policy claim.” This is a feature, not a constraint.

6.4. Election-Audit Analogy

Election certification does not rely on a single reported total. It reconciles several ledgers: eligible voters, ballots issued, ballots returned, precinct tabulations, canvass corrections, chain-of-custody records, and risk-limiting audit samples. The same design principle applies here.

For reapportionment stability, the audit package should reconcile:

1. the official apportionment count and HUNTINGTON-HILL calculation;

2. the graph and population-input hashes;
3. the split prescription for each state and chamber;
4. the deterministic seed or seed-family manifest;
5. the old and new assignments after canonical district matching;
6. the Hamming, compactness, split, and demographic deltas; and
7. any human-facing canvass note explaining why a scenario value differs from the final certified value.

This is also the natural bridge to precinct-vote auditing. A precinct vote audit should make its own reconciliation ledger explicit: registration eligibility universe, ballots issued, ballots cast, rejected/provisional ballots, tabulator totals, canvass adjustments, and sampled hand-count checks. The transferable principle is simple: every public total should have a lower-level manifest that lets an observer trace how the total was produced.

6.5. Implications for DIA Drafting

The analysis of this paper has three implications for the DIA statutory text:

1. **No grandfathering clause is needed.** Some proposals suggest that the DIA should include a “continuity bonus” rewarding plans that resemble the previous decade’s map. This paper suggests that the APPORTIONREGIONS algorithm can produce measurable reapportionment-stability records without an explicit continuity bonus. A continuity bonus would reduce compactness without providing meaningful stability gains.
2. **No transition period is needed for algorithm output.** The new maps are immediately valid after the reapportionment. The 23% Hamming distance for the Texas scenario should be treated as an auditable disruption estimate, not as a crisis requiring legislative remediation by itself.
3. **The prime factorisation changes are predictable.** States and administrators can pre-compute scenario tree structures from projection inputs years before the census. This allows for early public comment on the expected map changes before the official redistricting cycle begins.

6.6. A Note on the 2030 Texas Case

Texas going from $k = 38$ to $k = 41$ (prime) is the most structurally disruptive reapportionment in the projection set. A binary 20/21 fallback for Texas is a qualitatively different map from a largest-prime-first 19-container partition. The two maps will look different to voters, even if the compactness is similar.

This is not a failure of the algorithm; it is the correct response to the arithmetic of apportionment. Under the current implementation, 41 triggers the large-prime binary fallback. There may be other policy choices for $k = 41$, but they would be different algorithms and should be named as such. Observers should understand this before comparing the scenario to nearby counterfactuals such as $k = 40 = 2^3 \times 5$, which would produce a different four-level tree.

7. Conclusion

7.1. Summary

We have characterised the 2030 reapportionment shock for the APPORTIONREGIONS bisection algorithm. Using a 2030 projection scenario and simulation targets on 2020 census-tract graphs, we define the following scenario findings:

1. **Texas is the most disruptive case.** $38 = 2 \times 19 \rightarrow 41$ (prime) changes the tree from 19 two-seat containers to a binary 20/21 fallback. Scenario Hamming distance: $d_{\text{Ham}} = 0.23$.
2. **Florida is also dramatically affected.** $28 = 4 \times 7 \rightarrow 29$ (prime) produces a similar change from a seven-container top level to a binary 14/15 fallback. Scenario $d_{\text{Ham}} = 0.19$.
3. **Scenario disruption is far below full MCMC mixing.** The worst-case reapportionment disruption ($d_{\text{Ham}} = 0.23$) is equivalent to approximately 10 one-step RECOM units for Texas under the $1/k$ scale approximation. A fully-mixed ensemble (10,000+ steps) produces $d_{\text{Ham}} \approx 1$.
4. **Political redistricting is more disruptive.** The draft comparison range for contested

political redistricting cycles is 35–55% of tract assignments, 1.5–2.4× more than the Texas scenario. This comparison still needs same-metric evidence.

5. **Fresh recomputation is the correct statutory design.** The DIA’s decadal recomputation produces maps optimal for the current apportionment and geography. No continuity bonus or grandfathering clause is required by the current algorithmic staging; adding one would be a separate policy choice.

7.2. The Arithmetic of Disruption

The key insight is that reapportionment disruption is determined by two factors: seat count change and factorisation change. A state gaining one seat from a composite k to another composite $k + 1$ (e.g., Georgia $14 \rightarrow 15$, both composite) has disruption $d_{\text{Ham}} \approx 0.08$. A state moving from a composite largest-prime-first top level to a large-prime binary fallback (Texas, Florida) has scenario disruption $d_{\text{Ham}} \approx 0.20$ – 0.23 .

This arithmetic is predictable and transparent. States and administrators can compute scenario disruption years in advance from projection inputs, then replace the scenario with certified census data when the official apportionment is available.

7.3. Open Questions

1. **2030 census data.** The simulations use 2020 data with new seat counts. The actual 2030 disruption will be larger due to population redistribution. A 2030 simulation requires certified 2030 census inputs and final apportionment values.
2. **Optimal transition strategies.** If continuity is valued, is there an APPORTIONREGIONS variant that would minimise d_{Ham} between the old and new maps subject to compactness? This is an open algorithmic question.
3. **The prime frequency problem.** Among the 50-state seat counts, primes appear frequently (current primes: $\{7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$). A theory of reapportionment stability that accounts for prime density in the relevant range $[1, 60]$ would be useful. The prime number theorem gives density $\sim 1/\ln(k)$, so roughly $1/\ln(40) \approx 27\%$ of seat counts near 40 are prime.

7.4. Practical Advice

For states expecting seat changes in 2030, we recommend:

1. Pre-compute the 2030 bisection tree from current projections.
2. Run a preliminary simulation using 2020 data with the projected seat count to estimate boundary disruption.
3. Publish the projected changes for public comment before the 2030 redistricting cycle begins.

This transparency is consistent with the DIA’s overall commitment to deterministic, auditable redistricting.

References

- Michel L. Balinski and H. Peyton Young. Fair representation: Meeting the ideal of one man, one vote. *Yale University Press*, 1982. Comprehensive treatment of apportionment methods. Chapter 4 covers Huntington-Hill and its properties. Seat allocation stability under population change.
- Gio Della-Libera. ApportionRegions: Prime-factorisation bisection trees linked to Huntington-Hill apportionment, 2026. T.4 in Algorithmic Redistricting series. Key result: zero seed variance for tree structure. Tree topology is determined by the split prescription for k : largest-prime-first for composites and binary fallback for large primes. T.9 extends this by asking what happens when k changes at reapportionment.
- Edward V. Huntington. The apportionment of representatives in congress. *Transactions of the American Mathematical Society*, 30(1):85–110, 1928. doi: 10.2307/1989268. Original Huntington-Hill equal proportions method. The method of equal proportions is the basis for the HUNTINGTON-HILL apportionment used in APPORTIONREGIONS.
- U.S. Census Bureau. Projected congressional apportionment: 2030 projections. Technical report, U.S. Census Bureau, Population Division, 2025. Working projections for 2030 apportionment. TX: 38 $\rightarrow$ 41; CA: 52 $\rightarrow$ 50; FL: 28 $\rightarrow$ 29; NY: 26 $\rightarrow$ 25. Subject to revision pending final 2030 census enumeration.

Wesberry v. Sanders. Wesberry v. sanders, 1964. 376 U.S. 1 (1964). Equal population requirement. Motivates the decadal fresh recomputation of the APPORTIONREGIONS tree from the new apportionment — the intended DIA behavior.